

Effects of varying pH on the growth and physiology of five marine microphytobenthic diatoms isolated from the Solthörn tidal flat (southern North Sea, Germany)

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ABSTRACT: Diatoms inhabiting intertidal flats are subject to strongly changing environmental conditions, which have a great influence on the oxygen, sulphide and pH gradients in the sediments. In previous studies, variations in pH in the range from 6.5 to 8.5 had only minor effects on the growth of benthic diatoms. In order to determine the physiological responses of marine benthic diatoms to pH changes, cultures of *Amphora graeffii*, *Navicula gregaria*, *Navicula phyllepta*, *Nitzschia epithemoides* and *Pinnularia ambigua* (= *Biremis ambigua*) isolated from the Solthörn tidal flat (lower Saxony, southern North Sea) were used to study the effect on growth rates and biochemical compositions at different pH values (7.25, 7.5, 7.75, 8.0, 8.25, 8.5 and 8.75). During short- (6 h) and long-term exposures (30 d), the composition of free amino acids and the accumulation of polyols, saccharides, quaternary ammonium compounds and β -dimethylsulphoniopropionate (DMSP) were determined. While growth rates of the five tested diatom taxa during short-term exposure to experimental pH were only hardly affected, long-term exposure to experimental pH led to decreasing growth rates in all tested species. Significant differences were found in physiological responses to pH treatments of the species tested. Growth in terms of chlorophyll *a* and protein contents showed decreasing values also after short-term exposure to experimental pH. Furthermore, at experimental short-term pH treatments, species accumulated higher levels of proline, glycerol, galactose and mannose, while only three out of the five tested taxa also accumulated high amounts of the quaternary ammonium compounds homarine and glycine betaine as well as DMSP. The physiological responses to long-term exposure to experimental pH showed distinct differences in the accumulation of proline, serine, taurine, glycerol and other compatible solutes. In conclusion, significant differences in osmolyte compositions were found in the five diatom species exposed to different pHs, suggesting specific intracellular acclimation processes. These responses provide a possible explanation for the insensitivity to short-term pH variations.

KEY WORDS: Amino acids, Carbohydrates, Compatible solutes, DMSP, Marine benthic diatoms, pH acclimatisation, Proline, Quaternary ammonium compounds, Tidal flats

INTRODUCTION

Intertidal mudflats represent a highly unstable environment, due to continuously changing water levels and currents, as well as temporary exposure to air (Reineck 1983). These habitats often harbour extensive communities of benthic diatoms, which are responsible for a major part of the primary production (Underwood & Kromkamp 1999). Such communities are mainly composed of multi-layered biofilms of epipelagic species, which may perform vertical migration (Underwood & Kromkamp 1999). These migrations are reduced by the thickness of the photosynthetic layer of intertidal sediments (2–4 mm: Jørgensen *et al.* 1983; Kühl *et al.* 1996; Garcia-Pichel *et al.* 1999).

Equilibration of seawater with rising CO₂ in the atmosphere has already lowered the ocean pH by 0.12 units compared to pre-industrial values, which in turn has increased the concentration of hydrogen ions by 30% (Houghton *et al.* 2001; Sabine *et al.* 2004; Raven *et al.* 2005). In coastal waters of the southern North Sea, significant pH decreases have been observed since 1987 (Provoost *et al.* 2010). Although the decline in pH is small

and recent for the Western Scheldt and the Wadden Sea (Provoost *et al.* 2010), it must be stated that these areas are very dynamic habitats with diel and seasonal fluctuations in environmental conditions, which have a great influence on the oxygen, sulfide and pH gradients in the sediments (Brinkhoff *et al.* 1998). In the absence of photosynthesis, the decrease in the sediment pH is caused by oxidation of organic material, and minimum values in the pH profiles correspond to depletion of oxygen and anaerobic respiration (Brinkhoff *et al.* 1998).

The effects of alterations in pH on marine coastal and open ocean phytoplankton species were studied under laboratory conditions by several authors, showing the optimum pH for growth is between pH 6.3 and 10 [e.g. Taraldsvik & Mykkestad 2000 for *Skeletonema costatum* (Greville) Cleve (= *Skeletonema marinoi* D. Sarno & A. Zingone)]. Furthermore, Hinga (2002) reviewed 21 studies on the effects of pH on marine phytoplankton and stated that some species can grow well at a wide range of pH (e.g. *Phaeodactylum tricornutum* Bohlin, Humphrey 1975), while others have growth rates that vary greatly over a 0.5 to 1 pH unit change (e.g. *Prorocentrum micans* Ehrenberg, Barker 1935). In contrast, studies dealing with marine benthic diatoms are sparse. Wang *et al.* (1998) showed that *Amphora coffeaeformis* (C. Agardh) Kützinger [= *Halammphora coffeaeformis* (C. Agardh) Mereschkowsky], *Navicula corymbosa* (C. Agardh) Cleve, *N. mollis* (W. Smith) Cleve and *Cocconeis*

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scutellum Ehrenberg were rather insensitive to changes in pH in a range from 6.5 to 8.5. At 0.5 pH units above or below their maximum growth rate, these species grew only 4% to 7% slower than at their maximum. At one pH unit above or below their maximum, their growth rate was only reduced by 12% to 37% (Wang *et al.* 1998).

Scholz & Liebezeit (2012a, b) observed, during a 1 yr investigation of the microphytobenthic community in the Solthörn tidal flat (southern North Sea), significant differences in the benthic diatom community compositions but none of the abiotic parameters determined during the annual cycle provided sufficient explanation for species occurrences. While the growth of several diatom taxa isolated from this area was only hardly affected by changes in salinities, temperatures and ultraviolet (UV) radiation under laboratory conditions, physiological acclimatisation processes to these fluctuating variables were found. The acclimatisation patterns indicated the ability of single taxa to tolerate such environmental conditions by changes in their biochemical composition, including the accumulation of so called 'compatible solutes' and changes in carbohydrate, fatty acid and amino acid composition (Scholz & Liebezeit 2012c, 2013; Scholz *et al.* 2014a, b). Compatible solutes include carbohydrates such as mannose, polyols, glycine betaine and β -dimethylsulphoniopropionate (DMSP) and the imino acid proline (Erdmann & Hagemann 2001). Particularly, proline appears to be the most widely distributed osmolyte accumulated under osmotic stress by algae (Kirst 1996; Plettner 2002; van Bergeijk *et al.* 2003; Krell *et al.* 2007). It is not yet fully understood how these compounds are involved in stress tolerance but it is assumed that their possible roles include the stabilisation of complex proteins and membranes and protection of the transcriptional and translational machinery (e.g. Chen & Murata 2002). Thus, in the present investigation, the effects of different pH values on five Solthörn tidal flat isolates were tested in order to (1) determine variations in growth (growth rates, chlorophyll *a*, protein contents); (2) identify changes in total carbohydrate and lipid concentrations as well as carbohydrate and amino acid compositions, with special emphasis on the proline concentrations and (3) test the ability of the strains to accumulate other compatible solutes, such as the quaternary ammonium compounds (QUARCS, glycine betaine, homarine) and/or DMSP.

MATERIAL AND METHODS

The diatoms *Amphora graeffii* Cleve, *Navicula gregaria* Donkin, *Navicula phyllepta* Kützing, *Nitzschia epithemoides* Grunow, and *Pinnularia ambigua* Cleve [= *Biremis ambigua* (Cleve) D.G. Mann] were isolated from the Solthörn tidal flat in summer 2008, when relatively high pH fluctuations (up to 1.14 pH units) on the sediment surface were recorded (Scholz & Liebezeit 2012b). Details about sampling stations, dates, isolation of species and initial cell volumes are given in Scholz & Liebezeit (2012d, e). Growth characteristics of the five isolates, such as temperature and pH optima under optimised culture conditions, are presented in Table 1.

If not otherwise mentioned, all chemicals used in this study were of the highest purity from Sigma/Aldrich Chemical Co. (St. Louis, Missouri USA). Axenic cultures of individual species were grown in 2 l Erlenmeyer flasks (approximately 500 mL culture volume) using f/2 enrichment of synthetic seawater [Tropic Marin® (Aquarientechnik GmbH, Wartenberg Germany) pH 8.04, salinity = 30]. The diatoms were purified from bacterial contaminants by spreading cells on 1.5% f/2 agar plates containing 5 $\mu\text{g mL}^{-1}$ tetracycline, 10 $\mu\text{g mL}^{-1}$ kanamycin and 20 $\mu\text{g mL}^{-1}$ ampicillin after a vortexing step of 60 s. The absence of bacterial contaminants was verified three different ways: (1) epifluorescence microscopy using the dye 4',6-diamidino-2-phenylindol (DAPI), (2) via cultivation on agar plates using BD Difco™ Marine Agar 2216 (Fisher Scientific GmbH, Schwerte Germany) and polymerase chain reaction according to the protocol given by Hewson *et al.* (2003). To reflect Solthörn tidal flat conditions (Scholz & Liebezeit 2012a, b), the medium was modified: (1) the N:P ratio was changed to 17:1 and (2) NaHCO_3 was added after autoclaving to a final concentration of 2 mmol l^{-1} . The photon flux rate was 600 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ (Philips Global Technology GmbH, Aachen Germany, Master TL-D 18W/840) with a light:dark regime of 16:8 h. Cultures were maintained at $18 \pm 0.5^\circ\text{C}$. Glass beads (Sartorius GmbH, Göttingen Germany, 300–700 μm in diameter) were placed on the bottom of culture vessels with a layer of approximately 3 mm thickness and used as substratum. Salinity, pH and conductivity were measured using handheld probes (YK-31SA, YK-2001PH SI Model 33, Engineered Systems and Designs-Model 600, Philips). Cultures were supplied with new medium every 7 d to maintain optimal growth. After biofilm development (2 wk after inoculation), the culture broths were removed with a sterile sleeve until only a thin liquid film remained over the biofilms. Subsequently, the experiments were started by addition of fresh culture medium.

Experimental design

Axenic isolates were cultivated in 2 l semi-continuous cultures under sterile conditions as described above. Tris (4.13 mmol l^{-1}) was added to the f/2 medium, which was then adjusted with 1 mol/L HCl and 1 mol/L NaOH to the following pH values: 7.25, 7.5, 7.75, 8.0, 8.25, 8.5 and 8.75 (± 0.01). Control and adjustment of pH was carried out twice per day during the experimental period. The first test series was the short-term experiment, running 6 h. The second test series, the long-term experiment, was concurrently started and run for 30 d. Six replicate flasks per treatment and species were used for conducting the experiment. Taxa were grown in biofilms on a quartz-glass bead layer in order to simulate environmental conditions. Once per day, 60 mL of culture suspension was replaced with fresh medium, which gave a dilution rate of 0.6 d^{-1} , to avoid nutrient depletion in the cultures. The position of replicate flasks was randomly changed every other day to eliminate any location effects within the incubator.

Cell counts were performed microscopically, using a Neubauer Improved counting chamber with 0.1 mm depth (LO Laboroptik GmbH, Lancing UK). Only cells that exhibited red fluorescence under UV light were counted. At

Table 1. Characteristics of the Solthörn tidal flat isolates during pH experiments as well as under optimised culture conditions, including culture medium, cell volume, temperature and pH optimum.

Taxon	Medium ¹	Temperature optimum (°C) ²	pH, optimum ²	Cell volume (µm ³) ³
<i>Amphora graeffii</i> Cleve	f/2	16 ± 4	7.9–8.6	2816 ± 194.9
<i>Navicula gregaria</i> Donkin	SWES	16 ± 3	7.9–8.3	982 ± 56.3
<i>Navicula phyllepta</i> Kützing	f/2	18 ± 3	8.0–8.3	183 ± 20.6
<i>Nitzschia epithemoides</i> Grunow	f/2	16 ± 6	7.8–8.2	1926 ± 135.7
<i>Pinnularia ambigua</i> Cleve	SWES	17 ± 4	7.9–8.6	816 ± 44.5

¹ Media used during growth optimisation experiments: f/2, after Guillard (1975); SWES, seawater medium (http://sagdb.uni-goettingen.de/culture_media/05%20Seawater%20Medium.pdf).

² Data obtained from pre-tests during growth optimisation experiments, using temperature and pH steps of 1°C and 0.1, respectively.

³ Data measured during growth optimisation experiments following Snoeijis *et al.* (2002).

least 1000 cells were counted in each sample at ×400 magnification. For cell volume determination, diatoms were embedded in Naphrax (refractive index 1.74; Northern Biological Supplies Ltd., Ipswich UK), and dimensions were measured under a light microscope at 1000× magnification. Cell volumes were calculated according to the method described by Snoeijis *et al.* (2002) and presented as µm³ in Table 1.

Generally, biofilms were allowed to grow without physical disturbance on the glass beads. Diatoms were removed for sampling by shaking on a rotating shaking device (350 rpm), dislodging the biofilm from the substrate and homogenising the culture. Microscopic observation of the beads after shaking confirmed the complete release of the cells. Biomass from the short-term (6 h) and the long-term experiments (30 d) was harvested by centrifugation at 680 × *g* for 10 min. Three replicates were pooled subdivided into 11 aliquots (0.5–1.0 g fresh weight), fixed in liquid nitrogen, lyophilised (Alpha 1–4 LSC lyophilisation system, Martin Christ, Osterode am Harz Germany) and used for different extraction procedures as described below.

Biochemical analysis

Chlorophyll was extracted with 90% acetone over 24 h in darkness at –80°C. Chlorophyll *a* was determined spectrophotometrically (Shimadzu UV–visible spectrophotometer UV-1603, Kyoto Japan) and corrected for phaeopigments, according to Lorenzen (1967). Protein was measured using the Lowry method as described by Herbert *et al.* (1971) using bovine serum albumin as a standard. Total soluble carbohydrates were determined using the phenol sulphuric acid method of Kochert (1978) and Ben-Amotz *et al.* (1985) incorporating the modifications of Merz (1994) and Buttery (2000). The samples were homogenized in 1 M H₂SO₄, and, after heating at 100°C for 60 min, 0.3 ml of the supernatant was transferred into a fresh vial and made up to 2 ml with deionized water. Sets of glucose standards were prepared, and 1 ml of 5% (w/v) phenol solution was added to all samples, after which, 5 ml of concentrated H₂SO₄ was added. Absorbance was read at 485 nm. The total lipid determination was based on the method of Bligh & Dyer (1959) as modified by Kates & Volcani (1966) and adapted by Merz (1994) and Buttery (2000) using methanol:chloroform:deionised water (2:1:0.8, vol:vol:vol). The extracts were dried under a stream of ultra-pure N₂ gas and placed in a vacuum desiccator over silica gel overnight and then weighed. Values are means (± standard error [SE]) and were first normalized per unit cell

volume (µg µm^{–3}) and then expressed as µg cell^{–1}. Analytical errors determined from threefold replicate analysis varied from 3.5% to 8.9%.

The extraction of polyols, carbohydrates, DMSP and quaternary ammonium compounds followed the method of Karsten *et al.* (1991), using 70% EtOH. Saccharides and polyols accumulated during the experiments were identified by thin layer chromatography (TLC Silica Gel 60 F254; Merck KGaA, Darmstadt Germany) following different methods. While mono- and disaccharides were identified and quantified by the method described by Zhang *et al.* (2009), polyols were quantified according to the method of Klaus & Rippahahn (1983). Briefly, samples (50 µL) were developed with a solvent system consisting of propan-2-ol/water (85:15, vol:vol). The developed plate was stained by spraying with anisaldehyde/H₂SO₄ reagent (Krebs *et al.* 1967) and heated at 110°C for 5 min. The TLC plate was scanned, digitized and analyzed by Un-Scan-It gel scanning software (Silk Scientific Corp., Orem, Utah, USA). As standards, arabinose, rhamnose, xylose, sorbose, sucrose, mannose, galactose, glucose, fucose, fructose, glycerol and erythritol were used. Since all tested saccharides and polyols were present in the samples, we decided to consider only compounds above a limiting concentration in the following analyses (20 fmol cell^{–1}), on the basis that a higher concentration level would equate to importance in the acclimation process.

Homarine (N-methyl picolinic acid) and glycine betaine were detected according to the method of Bregoff *et al.* (1953). Here a Reinecke fractionation was used as a first preparation step, where the reinackates without a free carboxyl group were precipitated first at a pH of 9.0. Afterward, the free betaines were obtained by acidification (pH 1.2) of the filtrate. The chromatography of 80 µL samples was then carried out in a ethanol–ammonia solvent system, consisting of ethanol (95%) and concentrated NH₄OH (95:5, vol:vol). As detection reagent Dragendorff solution (KBiI₄-reagent) was employed made according to Bregoff *et al.* (1953) without any modifications. This reagent gives orange–red to brown–purple spots on a yellow background and quantification was carried out by standard curve equations, using homarine and glycine betaine (Merck) as standard.

DMSP was separated on 0.25 mm silica gel G plates (Machery-Nagel, Düren Germany) developed with methanol:acetone:HCl (90:10:4, vol:vol:vol) (Summers *et al.* 1998) and visualized by Dragendorff's reagent according to the method of Awwad & Adelstein (1966). DMSP was estimated by area measurement of Dragendorff-positive TLC zones. The plots

Table 2. Concentrations (mean $\pm$ standard error) of carbohydrates (I–V) and amino acids (VI–XIV) for five diatom species at the start of the experiments. Quaternary ammonium compounds, glycerol and DMSP were not detected.

Compound	Species				
	<i>Amphora graeffii</i>	<i>Navicula gregaria</i>	<i>Navicula phyllepta</i>	<i>Nitzschia epithemoides</i>	<i>Pinnularia ambigua</i>
Carbohydrates					
I Arabinose (fmol cell ⁻¹)	12.8 $\pm$ 2.0	10.3 $\pm$ 2.1	11.6 $\pm$ 2.3	8.6 $\pm$ 1.1	8.7 $\pm$ 1.4
II Galactose (fmol cell ⁻¹)	50.4 $\pm$ 5.3	60.3 $\pm$ 4.7	48.3 $\pm$ 3.5	52.8 $\pm$ 6.3	44.1 $\pm$ 5.2
III Glucose (fmol cell ⁻¹)	65.1 $\pm$ 8.6	50.1 $\pm$ 6.3	51.5 $\pm$ 4.9	42.2 $\pm$ 2.7	67.9 $\pm$ 5.1
IV Mannose (fmol cell ⁻¹)	16.2 $\pm$ 3.9	24.2 $\pm$ 4.0	19.6 $\pm$ 5.1	14.1 $\pm$ 3.0	10.6 $\pm$ 1.9
V Other ¹ (fmol cell ⁻¹)	33.3 $\pm$ 4.2	50.8 $\pm$ 4.1	45.2 $\pm$ 4.6	36.8 $\pm$ 2.5	31.9 $\pm$ 4.4
Amino acids					
VI Alanine (fmol cell ⁻¹)	8.6 $\pm$ 0.8	2.9 $\pm$ 1.1	2.7 $\pm$ 0.9	3.3 $\pm$ 1.0	2.1 $\pm$ 0.7
VII Aspartic acid (fmol cell ⁻¹)	22.6 $\pm$ 2.3	19.3 $\pm$ 1.5	22.6 $\pm$ 3.1	16.5 $\pm$ 1.3	26.4 $\pm$ 3.3
VIII Lysine (fmol cell ⁻¹)	3.8 $\pm$ 0.5	4.2 $\pm$ 1.0	4.9 $\pm$ 0.7	3.7 $\pm$ 0.8	6.7 $\pm$ 1.5
IX Glutamic acid (fmol cell ⁻¹)	24.3 $\pm$ 2.9	14.6 $\pm$ 1.4	19.8 $\pm$ 2.0	20.3 $\pm$ 2.2	18.5 $\pm$ 1.9
X Proline (fmol cell ⁻¹)	2.5 $\pm$ 0.3	1.1 $\pm$ 0.2	2.0 $\pm$ 0.1	1.5 $\pm$ 0.1	2.3 $\pm$ 0.2
XI Serine (fmol cell ⁻¹)	3.0 $\pm$ 0.4	1.8 $\pm$ 0.6	2.4 $\pm$ 0.4	4.1 $\pm$ 0.9	3.3 $\pm$ 0.7
XII Taurine (fmol cell ⁻¹)	3.8 $\pm$ 0.6	2.2 $\pm$ 0.7	1.9 $\pm$ 0.6	2.8 $\pm$ 0.4	5.0 $\pm$ 1.0
XIII Valine (fmol cell ⁻¹)	2.1 $\pm$ 0.4	2.7 $\pm$ 0.3	2.7 $\pm$ 0.5	3.0 $\pm$ 0.4	2.9 $\pm$ 0.6
XIV Other ² (fmol cell ⁻¹)	32.4 $\pm$ 3.9	31.9 $\pm$ 2.3	35.8 $\pm$ 4.1	30.6 $\pm$ 5.2	33.8 $\pm$ 3.4

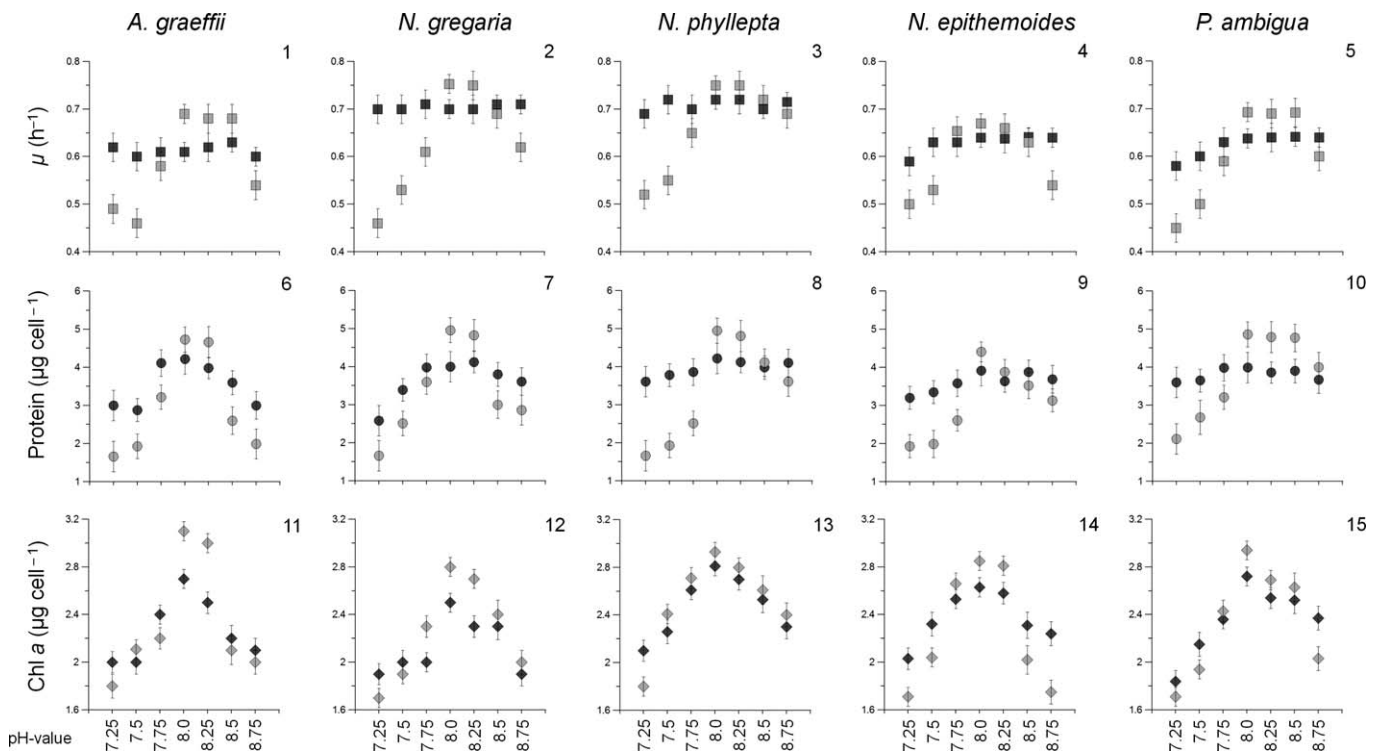
¹ Other: the sum of rhamnose, xylose, sorbose, sucrose, fucose, fructose.

² Other: the sum of arginine, cysteine, glycine, histidine, isoleucine, leucine, methionine, ornithine, phenylalanine, threonine, tryptophan, tyrosine.

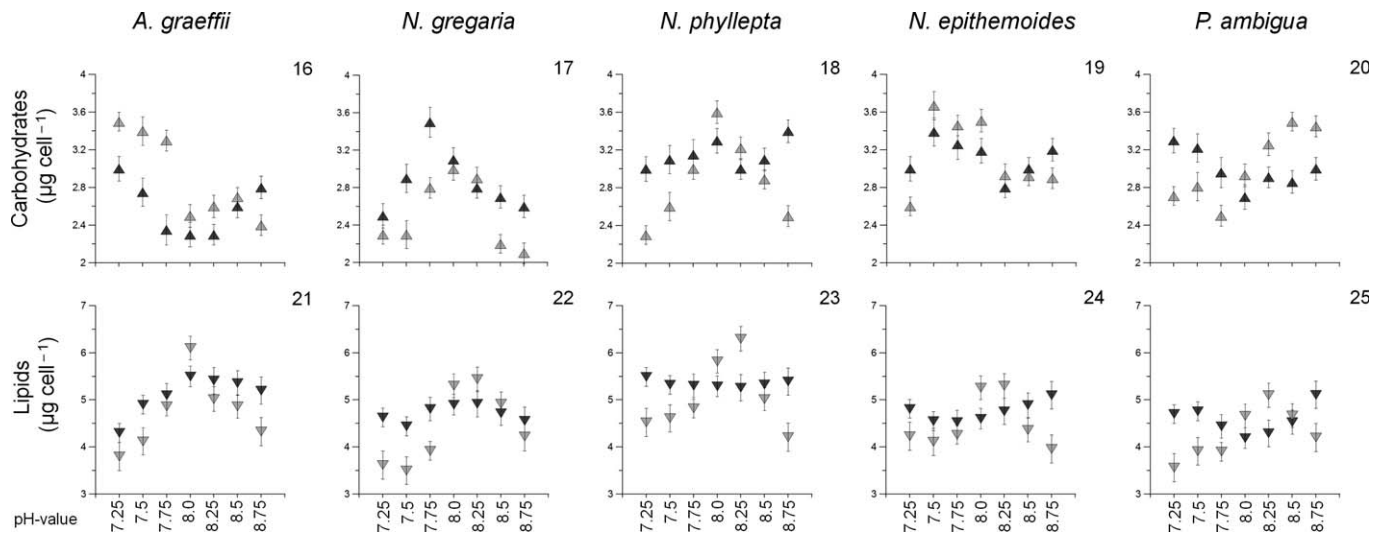
of zone area vs the logarithm of the quantity of standard (Fisher Scientific) were linear ($r^2 \geq 0.96$, $P > 0.01$). Values are means ($\pm$ SE) and were first normalized per unit cell volume (fmol μm^{-3}) and then expressed as fmol cell⁻¹ (Table 2) or relative concentrations (%; Fig. 3). Analytical errors determined from threefold replicate analysis varied from 5.4% to 11.9%.

Proline concentration was determined and quantified spectrophotometrically with ninhydrine, according to the

method described by Bates *et al.* (1973) and modified by Nothnagel (1995). For free amino acid analysis by high performance liquid chromatography (HPLC), cold perchloric acid extraction was applied according to the method of Bligny *et al.* (1989) and modified by Plettner (2002). The supernatant from the extraction was removed, nitrogen fixed and lyophilised (Alpha 1–4 LSC lyophilisation system, Martin Christ). The samples were stored at -80°C until analysis. HPLC analysis involved precolumn derivatisation with *o*-



Figs 1–15. Temporal variations in the specific growth rate (μ , Figs 1–5, $\square$), cell protein (Figs 6–10, $\circ$) and chlorophyll (Chl) *a* concentrations (Figs 11–15, $\diamond$) for *Amphora graeffii*, *Navicula gregaria*, *Navicula phyllepta*, *Nitzschia epithemoides* and *Pinnularia ambigua* at pH values of 7.25, 7.5, 7.75, 8.0, 8.25, 8.5 and 8.75 through short- (24 h, dark grey icons) and long-term experiments (30 d, light grey icons).



Figs 16–25. Temporal variations in total carbohydrates (Figs 16–20, Δ) and lipid concentrations (Figs 21–25, ∇) for *Amphora graeffii*, *Navicula gregaria*, *Navicula phyllepta*, *Nitzschia epithemoides* and *Pinnularia ambigua* at pH values of 7.25, 7.5, 7.75, 8.0, 8.25, 8.5 and 8.75 through short- (24 h, dark grey icons) and long-term experiments (30 d, light grey icons).

phthaldialdehyde (OPA) and 2-mercaptoethanol according to the method of Liebezit & Behrends (1999), using a 250 mm $\times$ 4.0 mm internal diameter column Nucleosil 100 C₁₈ with 5 μ m particle diameters as stationary phase (AB, Bischoff Chromatography, Leonberg, Germany). In detail, 1 ml sample or standard and 100 μ l sodium borate buffer (0.1 M, pH 10.5) were mixed for 50 s on a vortex. Then 50 μ l of the OPA reagent (prepared daily fresh by dissolving 20 mg OPA and 200 μ l 2-mercaptoethanol in 5 ml methanol) were added and mixed. After 2 min, 100 μ l of this mixture was manually injected with a glass syringe (Hamilton 710, Rheodyne 8125, Wertheim-Mondfeld, Germany). The column was eluted using a ternary gradient consisting of solvent A (0.05 M sodium acetate, pH 8.4), solvent B (0.05 M sodium acetate, pH 5.4) and solvent C (100% MeOH) as follows: 0–70% B and 20–30% C over 13 min, then 70–56% B and 30–36% C in 9 min, 100% C in 28 min, back to 0% B and 20% C in 5 min and maintaining 0% B and 20% C for 5 min. The flow rate was 1.0 ml/min. The amino acids were detected fluorometrically at an excitation wavelength of 330 nm and an emission wavelength of 450 nm. For calibration and quantification of amino acids a standard mixture of 18 amino acids (AA-S-18, Sigma) as well as L-ornithine and L-taurine were used. Quantification was carried out with the program EZChrom Elite 3.1.7 (Scientific Software Inc., Agilent Technologies, Waldbronn, Germany). Values are means ($\pm$ SE) and were first normalized to total cell volume (fmol μ m⁻³) and then expressed as fmol cell⁻¹ (Table 2) or relative concentrations (%; Figs 36–45). Analytical errors determined from threefold replicate analysis varied from 2.6% to 5.3%.

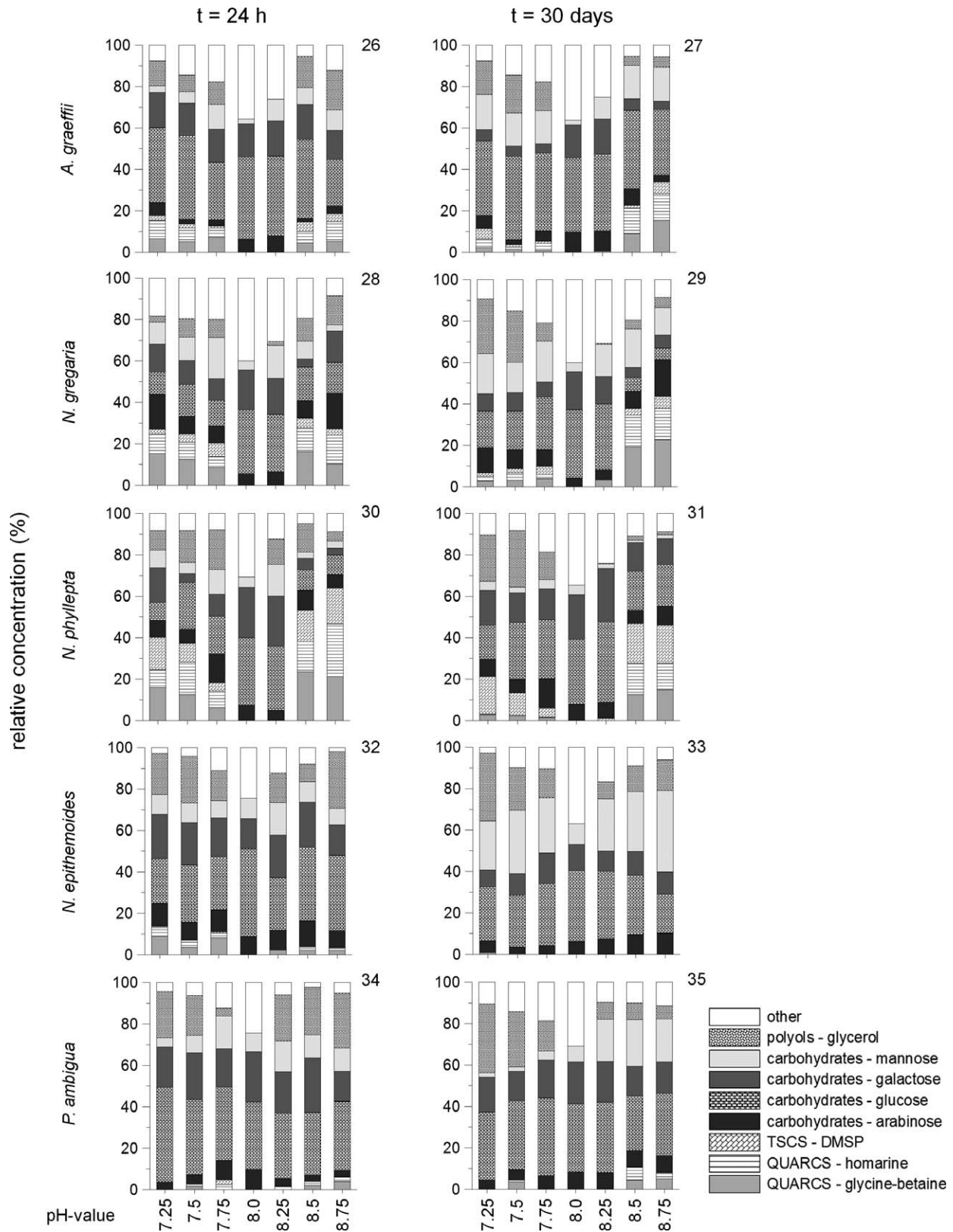
For all measured parameters, effects of the experimental pH treatments were statistically analysed with XLSTAT 2011, version 2011.2.08 (Addinsoft 2011). Effects of high pH (8.25, 8.5, 8.75) vs low pH (7.25, 7.5, 7.75) in comparison with the control (8.0) were tested with one-factorial analyses of variance (ANOVAs); *P* values of < 0.05 were accepted as representing significant differences.

RESULTS

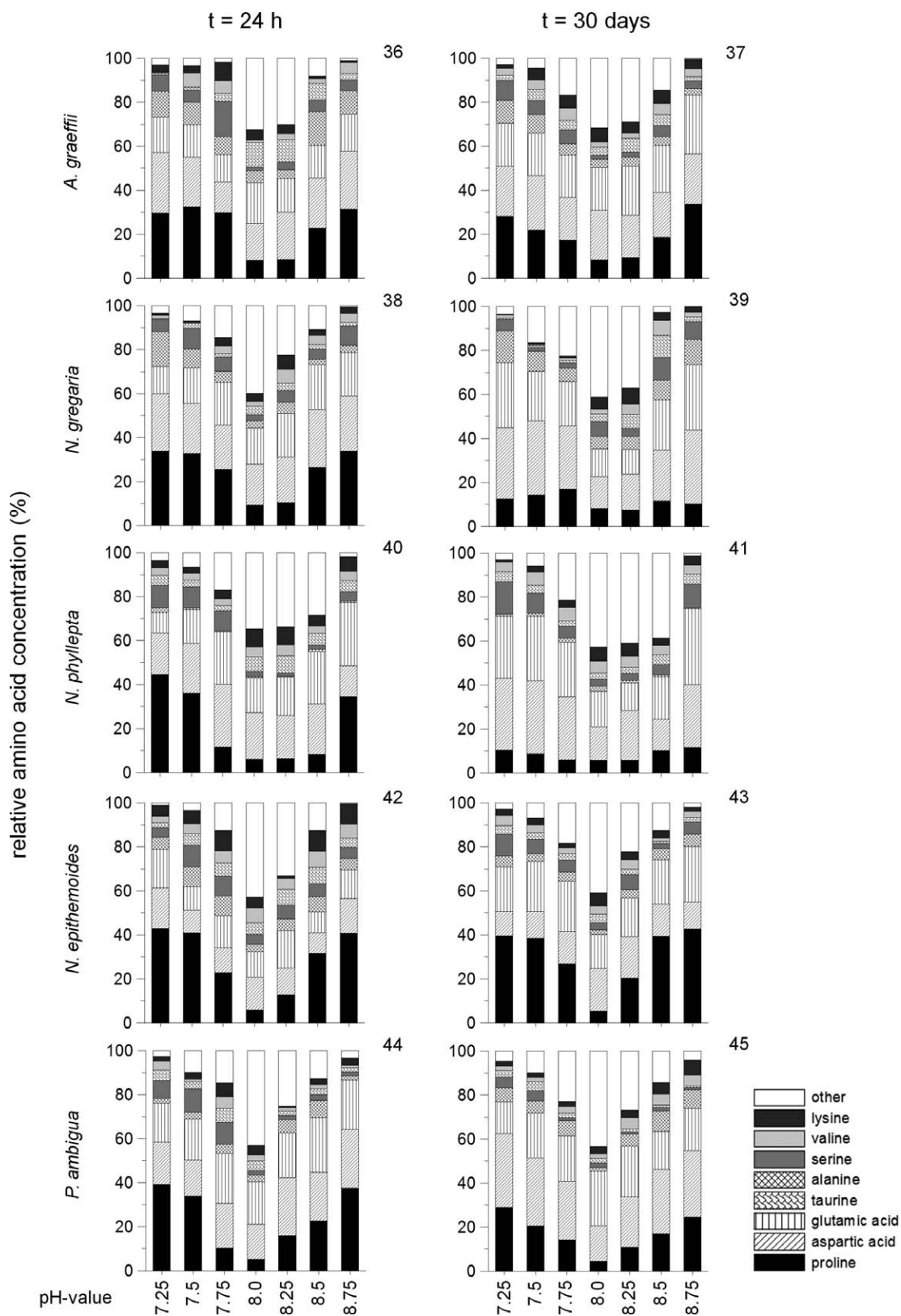
The variations in protein and chlorophyll *a* (Chl *a*) per cell, as well as growth rates based on cell numbers, were used to monitor the growth responses of the five benthic diatom taxa (Figs 1–15). With the exception of reduced cell numbers of *Navicula phyllepta*, *Nitzschia epithemoides* and *Pinnularia ambigua* in the lowest pH treatment (7.25), the relative growth rates of diatom cells grown at different pH values (7.5, 7.75, 8.0, 8.25, 8.5 and 8.75) varied only slightly during the short-term experiments ranging between 0.3% and 0.9% (ANOVA $F_{1,27} = 4.3$, $P = 0.45$, Figs 1–5). In contrast, long-term exposure resulted in decreasing cell numbers for most diatom species under experimental pH treatments (7.5, 7.75, 8.25, 8.5 and 8.75), being significant for *Na. gregaria* in the low-pH experiment (Fig. 2, difference between pH 7.25 and 8.0 = 31%, ANOVA $F_{1,26} = 27.2$, $P < 0.0001$), as well as *Ni. epithemoides* in the high-pH experiment (Fig. 4, difference between pH 8.0 and 8.75 = 33%, ANOVA $F_{1,26} = 21.9$, $P < 0.0001$). The control cultures, grown at a pH of 8.0, did not show significant differences in cellular growth rates over the duration of the experiments (average difference in cell density = 0.2–0.6%, respectively, $P > 0.05$).

In general, the cellular protein content was highly variable in short-term experiments and showed significant decreases in the long-term exposure to experimental pH treatments ($P < 0.05$, Figs 6–10). Particularly in *Navicula phyllepta*, the decrease in protein content in the low-pH assays was most notable (differences between the 24 h measurement and day 30 in the low- and high-pH treatment = 27–54%, ANOVA $F_{1,27} = 23.1$, $P < 0.0001$, respectively).

The cellular Chl *a* of all diatom species varied also with experimental pH (Figs 11–15). Species grown at experimental pH values had lower Chl *a* values than the 8.0 control treatment after short- and long-term exposure (average differences = 5–19%), being significant for *Nitzschia epithemoides* in both extreme pH assays (pH 7.25 and 8.75, ANOVA $F_{1,26} = 20.5$ and $F_{1,27} = 27.4$, $P < 0.0001$; Fig. 14)



Figs 26–35. Variations in relative QUARCS, DMSP, polyol and carbohydrate concentrations for *Amphora graeffii*, *Navicula gregaria*, *Navicula phyllepta*, *Nitzschia epithemoides* and *Pinnularia ambigua* at pH values of 7.25, 7.5, 7.75, 8.0, 8.25, 8.5 and 8.75 during 24 h (Figs 26, 28, 30, 32, 34) and 30 d (Figs 27, 29, 31, 33, 35) experiments.



Figs 36–45. Variations in relative amino acid concentrations for *Amphora graeffii*, *Navicula gregaria*, *Navicula phyllepta*, *Nitzschia epitnemoides* and *Pinnularia ambigua* at pH values of 7.25, 7.5, 7.75, 8.0, 8.25, 8.5 and 8.75 during 24 h (Figs 36, 38, 40, 42, 44) and 30 d (Figs 37, 39, 41, 43, 45) experiments.

and *Pinnularia ambigua* in the highest pH assay (pH 8.75, ANOVA $F_{1,27} = 30.6$, $P < 0.0001$; Fig. 15).

The cellular lipid and carbohydrate composition of each species showed significant differences between the individual diatom taxa in the control treatments (Figs 16–25). In detail, *Navicula phyllepta* had the highest carbohydrate content ($3.26 \mu\text{g cell}^{-1}$, ANOVA $F_{1,53} = 31.0$, $P < 0.0001$; Fig. 17); whereas, *Amphora graeffii* had higher cellular lipids ($5.51 \mu\text{g cell}^{-1}$, ANOVA $F_{1,53} = 25.5$, $P < 0.0001$; Fig. 21). Changes in cellular lipid and carbohydrate occurred across both pH treatment and exposure duration, showing highly variable patterns for the individual taxa. Increases of total carbohydrate content were observed for three out of the five tested diatom species after 24 h pH exposure, being significant for *Na. gregaria* in the 7.75 pH assay (ANOVA_{pH8.0:7.75} $F_{1,29} = 24.2$, $P < 0.0001$; Fig. 17) and *Na. phyllepta* in the highest pH treatment (ANOVA_{pH8.0:8.75} $F_{1,26} = 27.8$, $P < 0.0001$; Fig. 18). Similarly, total lipid concentrations increased in most species after short-term exposure to experimental pH values, being significant for *Pinnularia ambigua* in the highest pH treatment (ANOVA_{pH8.0:8.75} $F_{1,28} = 36.2$, $P < 0.0001$; Fig. 25). Furthermore, long-term exposure led, in most cases, to enhanced differences between control and treatments. Significant exceptions from this enhancement trend were the decreases of total carbohydrates for *A. graeffii* and *Na. phyllepta* after long-term exposure to a pH of 8.75 (ANOVA_{pH8.0:8.75} $F_{1,27} = 31.1$, $P < 0.0001$; Figs 16, 18), as well as the decreases of total lipid content for *Nitzschia epithemoides* in the high-pH assay (ANOVA_{pH8.0:8.75} $F_{1,27} = 26.1$, $P < 0.0001$; Fig. 24), and for *P. ambigua* in the lowest pH treatment (ANOVA_{pH8.0:7.25} $F_{1,29} = 30.7$, $P < 0.0001$; Fig. 25).

For a number of carbohydrates (rhamnose, xylose, sorbose, sucrose, fucose, fructose), both final concentration and variability across treatments was low, suggesting that these compounds were not involved in the pH acclimatisation processes. These carbohydrates are summed and reported as ‘other’ in Figures 26–35 and Table 2. They are not considered in detail for the following analyses.

With the exception of higher glucose content in *Pinnularia ambigua* (average diff.: 8.3%), the initial sugar composition of diatom taxa did not vary appreciably ($t = 0$ h, $P > 0.05$, Table 2). In three out of the five tested diatom taxa, glucose was the predominant sugar, with arabinose, mannose and rhamnose found in lower amounts with less variability between treatments (2.7–10%). Galactose (16.4–19.9%) was also common and was the predominant sugar in *Navicula gregaria* and *Nitzschia epithemoides*. QUARCS and the tertiary sulphonium compound DMSP were not detectable at the beginning of the experiments but were detectable and significantly different across species after 24 h (Figs 26–35). In detail, three out of the five tested diatom taxa accumulated higher proportions of carbohydrates and glycerol after 24 h pH exposure (up to 31.8%, ANOVA $F_{1,26} = 23.8$, 27.5 and 30.6, $P < 0.0001$, respectively; Figs 26, 32, 34); whereas, both Naviculaceae (*Na. gregaria* and *Na. phyllepta*) showed higher relative proportions of QUARCS and DMSP (ANOVA $F_{1,26} = 32.7$, $P < 0.0001$; Figs 28, 30). The highest relative content of glycine betaine and homarine were observed in *Na. phyllepta* after 24 h exposure to high pH values, ranging between 9.3–12.8% and 14.6–21.2% for

homarine and glycine betaine, respectively (Fig. 30). In general, accumulation of metabolites relative to controls occurred after 30 d of pH exposure. Glycerol accumulation was the main response to low-pH treatment in all tested species (Figs 27, 29, 31, 33, 35) and was significant for *Na. gregaria* and *Na. phyllepta* in the 7.25 pH assay (increases up to 26.8%, ANOVA $F_{1,27} = 25.1$, $P < 0.0001$, Figs 29, 31). Furthermore, high relative concentrations of QUARCS and DMSP were found in the high-pH treatments of *Amphora graeffii*, *Na. gregaria* and *Na. phyllepta* (Figs 27, 29, 31), being significant for *Na. gregaria* in the 8.75 pH assay (glycine betaine; ANOVA_{pH9.5(24h/30d)} $F_{1,27} = 28.3$, $P < 0.0001$; Fig. 31). In contrast, a significant decrease of glycine betaine, homarine and DMSP in *Na. phyllepta* in the highest pH assay after 30 d pH exposure was also recorded (ANOVA_{pH8.75(24h/30d)} $F_{1,27} = 21.9$, $P < 0.0001$; Fig. 21). Finally, only traces of glycine betaine were detected in the low-pH assays of *Ni. epithemoides* (0.11–0.29 fmol cell⁻¹ = 0.3–0.8%), while significant increases of mannose in the high-pH assays were found (up to 36.2%, ANOVA_{pH8.75(24h/30d)} $F_{1,27} = 29.7$, $P < 0.0001$; Fig. 33).

For the amino acids histidine, cystine, ornithine, arginine, tyrosine, phenylalanine, methionine, glycine, isoleucine and leucine, both final concentration and variability across treatments was low, suggesting that these compounds were not involved in the pH acclimatisation processes. These amino acids are summed and reported as ‘other’ in Figs 36–45 and Table 2.

With the exception of significantly higher values of aspartic acid in *Pinnularia ambigua* and alanine in *Amphora graeffii* compared with the other three isolates at $t = 0$ (ANOVA, $F_{1,27} = 22.3$ and 28.5, $P < 0.0001$, respectively; Table 2), the initial free amino acid compositions of the five benthic diatom taxa as well as in the pH controls during the experiments varied slightly, showing insignificant variations of serine, lysine and valine ($P > 0.05$). Differences occurred after 24 h exposure to high and low pH (Figs 36–45). The levels of free proline rose from initial values of 2.1–3.9 fmol cell⁻¹ in the control to 85.6–112.3 fmol cell⁻¹ in the high and low-pH assays, being significant for *Navicula phyllepta* at the lowest pH (ANOVA: $F_{1,27} = 30.1$, $P < 0.0001$; Fig. 40). In this context the significant increase of proline in the short-term high-pH assay of *Nitzschia epithemoides* is notable (ANOVA: $F_{1,26} = 27.3$, $P < 0.0001$; Fig. 42). Serine showed the highest relative concentration in the 7.5 treatment of *A. graeffii* (ANOVA: $F_{1,27} = 36.8$, $P < 0.0001$; Fig. 36); whereas, in the lower pH treatments values ranged from 5.4% (pH = 7.25, *Ni. epithemoides*, 19.6 fmol cell⁻¹; Fig. 42) to 12.8% (pH=7.5, *Pinnularia ambigua*, 27.3 fmol cell⁻¹; Fig. 44) of the total relative amino acid concentrations (ANOVA: $F_{2,26} = 30.3$, $P < 0.0001$). In addition, there were significant differences in the proportions of lysine (4.3–10.5%, ANOVA $F_{1,27} = 20.8$, $P < 0.0001$), taurine (0.9–5.7%, ANOVA $F_{1,27} = 23.9$, $P < 0.0001$) and alanine (2.3–11.3%, ANOVA $F_{1,27} = 33.2$, $P < 0.0001$) after short-term exposure to high pH. After 30 d of exposure, the pH induced levels of free proline decreased in three out of the five tested species ($t = 30$ d, Figs 37, 39, 41, 43, 45); whereas, the proline levels in *A. graeffii* and *Ni. epithemoides* remained high (up to 42.1% of the total relative amino acid concentrations at pH 8.75; Fig. 43). Furthermore, the high relative concentrations of taurine in

four out of the five tested taxa after 30 d are noteworthy, being significant for *P. ambigua* in both high-pH assays (ANOVA: $F_{1,27} = 28.3$ and 25.4 , $P < 0.0001$, respectively; Fig. 45).

DISCUSSION

Although the overall growth rates (μ) measured in this study are within the range of published values (Admiraal 1984; Peletier 1996; Underwood & Smith 1998; Smith 1999) for benthic marine diatoms, the pH optima described by several authors differ significantly from that found in the present study, suggesting species specific pH optima and ranges. For example, Taraldsvik & Mykkestad (2000) recorded for the marine phytoplankton diatom *Skeletonema costatum* a nearly constant growth rate at pH 6.5–8.5; whereas, Griffis & Chapman (1990) found the diatom *Ditylum brightwellii* (T. West) Grunow grew faster at pH 7.6–7.8 than at pH 8.2. Furthermore, Blanchemain *et al.* (1994) and Hinga (1992) found that diatoms and prymnesiophytes exhibit pH optima near 8.1 and clearly lower growth rates at pH 8.5. Goldman *et al.* (1982) stated that many marine species grow optimally at pH 8.1–8.3. The data in the present study suggested the ability of the five tested diatom taxa to grow under a wide pH range, with distinct differences between short-term and long-term exposure to experimental pH. Here, short-term growth rates, calculated for a period of 24 h, showed constant growth over a pH range from 8.0 to 8.5 for the majority of the tested diatom taxa, while long-term exposure to lower and higher pH led to decreasing growth rates in all tested species. In contrast, growth measured in terms of Chl *a* and proteins showed also clear effects (declining values) for the short-term treatments; whereas, long-term data were positively correlated with decreasing growth rates ($P < 0.05$).

Accumulation of intracellular, photosynthetic products (lipids or carbohydrates) is a common, although not unique, property of stressed algae (e.g. Harrison *et al.* 1990; De Madariaga & Joint 1992). In the present study, carbohydrate and lipid contents seemed to respond to pH, while in the control treatment, the biochemical compositions of the five diatom taxa were similar to that published for other diatoms (e.g. Rousch *et al.* 2003; Pistocchi *et al.* 2005). In several cases, the increase of proteins and carbohydrates were correlated to decreasing Chl *a* concentrations in the five tested diatom taxa. Karentz *et al.* (1991) stated that a general increase in cell dry weight of stressed algae reflects a storage of proteins and carbohydrates due to delayed cell division. Thus the biochemical data reflect a narrower pH optimum than observed for growth rates.

The carbohydrate analysis in the present study did not allow the differentiation between intra- and extracellular monosaccharides. Consequently, the increased carbohydrate contents might be also derived from the exopolysaccharide (EPS) fractions of the tested diatom taxa. Generally, the excretion of EPS can serve as a boundary between cells and the surrounding environment and fulfils a protective role against desiccation (De Caiola *et al.* 1996; De Philippis *et al.* 2001), and it may serve for heterotrophic metabolism in

algae in surviving prolonged periods under detrimental conditions (Gross *et al.* 1998).

Besides the free amino acids, several other organic osmolytes have been detected in marine and estuarine diatoms, such as glycine betaine (Dickson & Kirst 1987), homarine (Dickson & Kirst 1987; Nothnagel 1995), cyclohexanetetrol (Fujii *et al.* 1995; Garza-Sánchez *et al.* 2009), glycerol (Dickson & Kirst 1987) and finally, mannose (Paul 1979). While glycerol and mannose are direct products of photosynthesis, polyols (e.g. glycerol) and heterosides, glycine betaine, homarine and cyclohexanetetrol are not. Studies *in vitro* suggest that the polyols could serve as compatible solutes, acting as low-molecular-weight chaperones and as oxygen radical scavengers (Parida & Das 2005). In diatoms, the only reported carbohydrates and polyols accumulating as a response to salinity increases are mannose in *Cylindrotheca fusiformis* B.E.F. Reimann & J.C. Lewin, 1,2,4,5 cyclohexanetetrol in *Chaetoceros muelleri* Lemmermann, and glycerol in *Phaeodactylum tricornutum* (Paul 1979; Dickson & Kirst 1987; Fujii *et al.* 1995). In the present study, the accumulation of glycerol was the main response to both lower and higher pH in two out of the five tested diatom taxa in the short-term experiments. In long-term exposure, glycerol accumulated predominantly in low-pH treatments. In contrast, two out of the tested species accumulated higher amounts of homarine and glycine betaine as a response to experimental pH in both long- and short-term experiments, while DMSP was only found in one species in higher relative concentrations. Generally, not all organic solutes have equal protective properties. Based on a comparison of the general properties of compatible solutes and *in vitro* enzyme assays, Kirst (1996) concluded that DMSP might be less effective as compatible solute than, e.g. betaine, proline or glycerol. For instance, DMSP does not seem to be synthesized rapidly following changes in water potential. In the present study, most observations indicated that under such conditions DMSP is produced only slowly, if at all.

Additionally, glucose accounted for > 34% of the carbohydrate fraction, which is slightly higher than the 20–30% recorded in other studies (Hama 1988). The high values of glucose in the carbohydrate pool, however, does not seem to be uncommon. Hama & Handa (1992) have observed that glucose can account for 65–85% of the total monosaccharides in a coastal phytoplankton bloom dominated by diatoms. Percival *et al.* (1980), showed that p-1,3-D-glucan, the major storage carbohydrate in phytoplankton, yielded mainly glucose upon hydrolysis. In addition to glucose, mannose and galactose also contributed significantly to the overall increase in the monosaccharide content of the carbohydrate pool in the pH treated taxa in the present investigation. Here, mannose was predominantly detected in all five species as response to long-term pH exposures; although, distinct differences between species and exposure to low and high pH were distinguishable.

With the exception of the work of Taraldsvik & Mykkestad (2000), there appear to be no published data on cellular content and composition of amino acids in marine diatoms at different pH values. Generally, proline is considered to be involved in the osmotic adjustment of benthic marine diatoms (Scholz & Liebezeit 2012c). That is,

it acts as a compatible osmolyte and functions as a protector of macromolecules such as proteins and membranes and as nitrogen-storage compound and energy source after stress release (Aspinall & Paleg 1981; Delauney & Verma 1993). Furthermore, proline accumulation has been reported during nitrate deficiency because it is directly derived from glutamic acid and is easily convertible to various other amino acids, including glutamic acid (Mifflin & Lea 1977). In the present study, the short-term accumulation process to experimental pH was accompanied by an increase in proline levels in all tested species; while after long-term exposure relative proline concentrations were significantly lower in three out of the five tested taxa. Similar results were obtained by Plettner (2002) and Krell *et al.* (2007) in relation to long-term acclimatisation effects to different salinities in Antarctic diatoms. The authors stated that the decrease in proline levels after long-term exposure might reflect a stabilisation of the acclimatisation process in the cell metabolism to changes in the experimental conditions. Indeed, the data in the present study suggest for both Naviculaceae that proline levels decline in response to long-term pH exposure.

Besides proline, several other free amino acids have been detected in marine diatoms with possible function as organic osmolytes such as glutamic acid (Fujii *et al.* 1995; Nothnagel 1995), alanine (Nothnagel 1995) or taurine (Jackson *et al.* 1992). All these compounds were described in relation to salinity stress (Fujii *et al.* 1995; Nothnagel 1995); whereas, the role of amino acids involved in pH acclimation processes in benthic diatoms apparently has not been considered so far. While several amino acids were present only in marginal concentrations, the five analysed diatom taxa showed significant variations of serine and taurine in relation to pH exposure. In the presence of different pH, the decrease in the relative concentrations of glutamic acid and aspartic acid, and the increase in taurine along with proline, represented the most conspicuous changes within the amino acid pools of the five diatom taxa.

Acclimation and adaptation

Generally, acidification of marine waters, caused by anthropogenic atmospheric CO₂ production, is a topic of concern because a decrease in the pH of surface waters has multiple effects on the functioning, productivity, growth and survival of marine organisms (e.g. Guinotte & Fabry 2008). While direct acidification effects have been shown mainly for marine calcifying organisms (e.g. Hoegh-Guldberg *et al.* 2007), indirect acidification effects can be expected for consumers by changing the nutritional quality of their prey (Rossoll *et al.* 2012). In tidal flats, resuspended microphytobenthos often exceed the phytoplankton biomass in the water column (e.g. de Jonge & van Beusekom 1992). Consequently microphytobenthos, and in particular benthic diatoms, become an important diet of benthic suspension feeders inhabiting tidal flats or tidal channels such as the blue mussel, *Mytilus edulis* Linnaeus (e.g. Riera & Richard 1996; Kang *et al.* 2003). Numerous feeding experiments conducted for aquaculture have demonstrated the importance of the biochemical composition of microalgae on health and reproduction of planktonic and benthic grazers (e.g. Enright *et al.* 1986; Wikfors *et al.* 1992; Leonardos & Lucas 2000). Particularly

long-chain polyunsaturated fatty acids, such as docosahexaenoic acid (C22:6 w-3, DHA), eicosapentaenoic acid (C20:5 w-3, EPA) and arachidonic acid, play an important role in growth, development and reproduction success in heterotrophs (Müller-Navarra *et al.* 2004; Glencross 2009). For example, copepods retain polyunsaturated fatty acids such as EPA and DHA at all stages during development (Graeve *et al.* 2005). Furthermore, changes in lipid, carbohydrate and protein content of unialgal diets have been shown to significantly correlate with the growth of bivalves such as the clam *Mercenaria mercenaria* Linnaeus (Wikfors *et al.* 1992) and juveniles of the European oyster *Ostrea edulis* Linnaeus (Enright *et al.* 1986). Besides the importance of biochemical gross composition, the effects of amino acid and monosaccharide concentrations and composition on the growth of bivalves have also been discussed (e.g. Webb & Chu 1983). However, species-specific requirements of suspension feeders have been reported by different authors (e.g. Brown *et al.* 1997). For example, algal diets rich in polyunsaturated fatty acids and with high levels of carbohydrate have been shown to produce the best growth for juvenile oysters (*O. edulis*; Enright *et al.* 1986) and larval scallops (*Patinopecten yessoensis* Jay; Whyte *et al.* 1989); whereas, high levels in protein provides best growth for juvenile mussels (*Mytilus trossulus* Gould; Kreeger & Langdon 1993). Although these findings are difficult to compare with field conditions, the changes in biochemical composition of some benthic diatom taxa in response to fluctuating pH values, as recorded in the present study, may give a first indication on indirect effects in the benthic food web caused by the acidification of marine waters. More detailed investigations are needed to observe patterns of pH caused changes in biochemical compositions of benthic diatom taxa and to relate these variations in a context to microphytobenthic food web dynamics.

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